

Timur Khairulov

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AutoML Lab | ml.cau.ac.kr | Supervisor: Professor Jaesung Lee

PUBLICATIONS

Journal Articles

1. ChainImputer: A Neural Network-Based Iterative Imputation Method Using Cumulative Features
MDPI Symmetry | 2025 | DOI: [10.3390/sym17060869](https://doi.org/10.3390/sym17060869)

Conference Papers

1. Diffusion Model-Based Generative Pipeline for Children Song Video
IEEE International Conference on Consumer Electronics (ICCE) | 2025 | [IEEE Xplore](#)
2. Review of current approaches in the area of object detection for autonomous vehicles
Institute of Electronics and Information Engineers (IEIE) | 2024 | [DBPIA](#)

Under Review

1. Information-Theoretic Unsupervised Feature Selection for Constructing Taxonomy Tree
2. BAGen: Background Animation Generation for LLM-Generated Lyric Towards Children's Songs
3. Efficient Multi-Scale Network for Real-Time Crack Segmentation: Achieving High Accuracy-Speed Trade-off
4. Efficient Cheap All-Layer Aggregation Network for Time-Sensitive Time Series Classification

Theses

1. **Master's Thesis:** A Study on Maximizing Joint Entropy and Pattern Discrimination for Unsupervised Feature Selection | *Expected 2025*
2. **Bachelor's Thesis:** [Recognition of road markings, signs and traffic lights using deep learning models](#) | 2023

EDUCATION

2024 - Present	MS in Artificial Intelligence at Chung-Ang University CAYSS scholarship recipient	(GPA: 4.22/4.5)
2018 - 2023	BS in Computer Science at Electrotechnical University "LETI"	(GPA: 4.3/5)
2022 - 2023	Exchange Semester at Inha University Certificate	(GPA: 3.9/4.5)

WORK EXPERIENCE

C++ Software Developer LTE L1, YADRO May 2023 - February 2024
– 4G Base Station Development, used AVX and C++ for developing real-time modules

TEACHING EXPERIENCE

Teaching Assistant, Computer Vision, Chung-Ang University

Spring 2025

- Prepared course materials including PowerPoint presentations, designed and graded examinations, evaluated student homework assignments

PROJECTS

VLM Compression: Optimization techniques for Vision-Language Models

[GitHub](#)

Python, PyTorch, VLMs | December 2025 | Quantization and structural pruning methods for VLMs (LLaVA, BLIP-2) with performance benchmarking

Curriculum Learning Fine-Tuning for Mathematical Reasoning

[GitHub](#)

Python, PyTorch, Transformers, LoRA | November 2025 | Curriculum learning strategies for fine-tuning LLMs on GSM8K dataset, achieving 1.37% improvement

BAGen: Automatic production and visualization of children songs

[GitHub](#)

Python, PyTorch | January 2025 | End-to-end generation of content for children with GUI and TUI support

Feature Selection experimental code

[GitHub](#)

Matlab | August 2025 | Collection of Unsupervised Feature Selection algorithms and experiments used in our recent publications

Prim's Algorithm Visualizer

[GitHub](#)

C++, SFML, Make | April 2023 | Interactive graph creation and Prim's algorithm visualization using computer graphics

RESEARCH INTERESTS

- Large Language Models (LLMs)
- Efficient AI and Model Compression
- Feature Selection and Information Theory
- Vision-Language Models

TECHNICAL SKILLS

Programming	C/C++, Python, MATLAB, Dart, SQL
Machine Learning	PyTorch, NumPy, Pandas, Scikit-learn, OpenCV
Development	Git, Linux, CMake, Docker, Qt, Flutter
Systems	Parallel Programming, Operating Systems, Computer Architecture
Languages	English (IH - OPIc Certificate), Russian (Native), Korean (Basic)

AWARDS AND HONORS

2024 - Present CAYSS Scholarship Recipient at **Chung-Ang University**

2018 - 2023 Bachelor's Degree Scholarship Recipient at **Electrotechnical University "LETI"**